

Name: \_\_\_\_\_

**Use the Base Angles Theorem and its converse.**

1. The vertex angle of an isosceles triangle measures  $52^\circ$ . Find each base angle.
2. A base angle of an isosceles triangle measures  $37^\circ$ . Find the vertex angle.
3. The base angles of a triangle measure  $(2x + 13)^\circ$  and  $(4x - 9)^\circ$ . Find  $x$  and each base angle.
4. Each angle of an equilateral triangle is written as  $(5y - 5)^\circ$ . Find  $y$ .

**Use the side lengths.**

5. The legs of an isosceles triangle measure  $3x + 4$  and  $5x - 12$ . Find  $x$  and the length of each leg.
6. That triangle has a base of 18. Find its perimeter.

**Answer each question.**

7. In  $\triangle DEF$ ,  $m\angle D = 68^\circ$  and  $m\angle E = 68^\circ$ . What can you conclude about the sides? Name the theorem.
8. A triangle has angles measuring  $60^\circ$ ,  $60^\circ$ , and  $60^\circ$ . Classify it two ways and justify.
9. In the diagram,  $\overline{AB} \cong \overline{AC}$ ,  $\overline{BD} \cong \overline{BC}$ , and  $m\angle A = 38^\circ$ . Find  $m\angle DBC$ .
10. Explain why the vertex angle of an isosceles triangle can never measure  $180^\circ$  or more.

